

EE3621: Digital Signal Processing

Lecture 25: Moving-Average Filters & Comparison of FIR vs. IIR Filters (III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Analyze** the frequency selectivity and pole-zero placement of Moving Average and Comb filters.
 2. **Compare** FIR and IIR digital filters across all architectural and algorithmic metrics.
 3. **Select** the optimal filter topology for specific application constraints (latency, phase linearity, computational budget).
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2. MATHEMATICAL FOUNDATIONS

2.1 Moving Average Filter

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] \implies H(z) = \frac{1 - z^{-M}}{M(1 - z^{-1})}$$

$$H(e^{j\omega}) = \frac{1}{M} \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j\omega(M-1)/2}$$

- $M - 1$ zeros are located on the unit circle at $z = e^{j2\pi k/M}$ for $k = 1, 2, \dots, M - 1$.
- The pole at $z = 1$ exactly cancels the zero at $z = 1$.

2.2 Comprehensive FIR vs. IIR Comparison Matrix

— Engineering Metric — FIR Filter — IIR Filter — — : — — : — — — **Phase Response** — Can guarantee **strictly linear phase** ($\tau_g = \text{const}$) — Inherently **non-linear phase** (phase distortion) — — **Stability** — **Always BIBO stable** (All poles at origin $z = 0$) — Must be tested; poles can migrate outside $|z| = 1$ — — **Filter Order** — **High order** required for sharp transition bands — **Low order** ($5\times$ to $10\times$ lower than FIR) — — **Computational Cost** — High MAC count per sample — Low MAC count (extremely efficient) — — **Hardware Realization** — Non-recursive (no feedback loops); pipelineable — Recursive (feedback loops); accumulator bottlenecks — — **Quantization Effects** — Low sensitivity; free from

feedback limit cycles — High sensitivity; susceptible to limit cycles & overflow — — **Analog Emulation** — Cannot easily emulate analog prototypes — Directly designed from analog Butterworth/Chebyshev —
Pictorial Representations

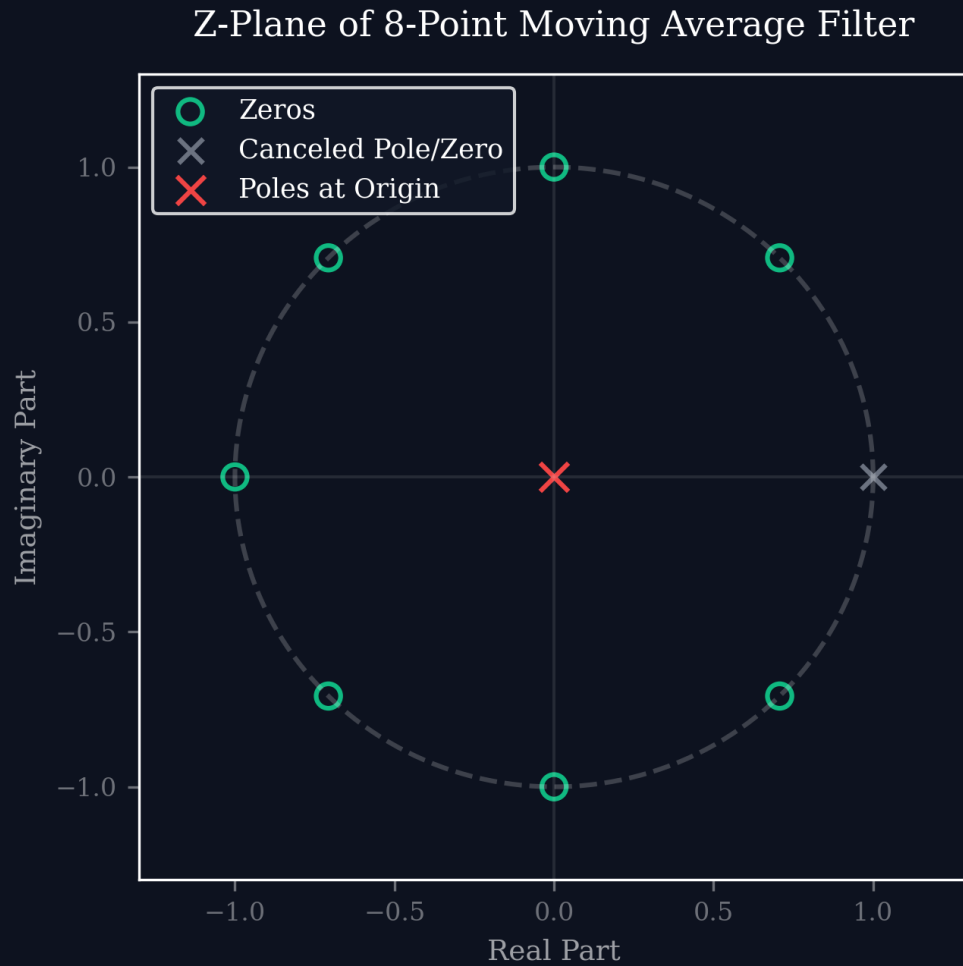


Figure 1: Moving Average Zplane

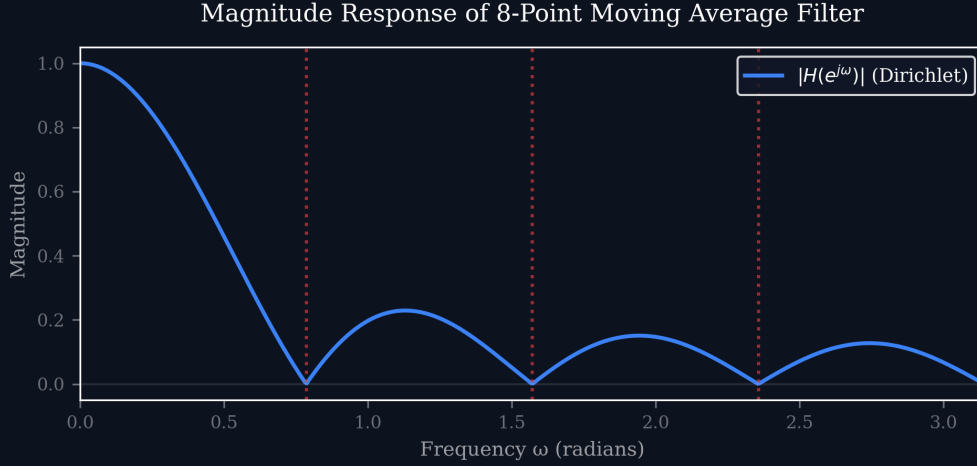


Figure 2: Moving Average Frequency

3. WORKED NUMERICAL EXAMPLES

Example 25.1: Order Estimation Comparison (FIR vs. IIR)

Problem: Estimate the required filter order for an FIR filter (Kaiser) vs. an IIR filter (Butterworth) to meet: $\omega_p = 0.4\pi$, $\omega_s = 0.5\pi$, $A_p = 0.5$ dB, $A_s = 50$ dB.

Solution:

- **FIR (Kaiser Formula):**

$$\Delta f = \frac{0.5\pi - 0.4\pi}{2\pi} = 0.05. \quad N_{\text{FIR}} \approx \frac{50 - 7.95}{14.36 \times 0.05} + 1 = \frac{42.05}{0.718} + 1 = 58.56 + 1 \approx \mathbf{60} \text{ taps.}$$

- **IIR (Butterworth Order Formula):**

$$\text{Prewarping / prototype ratio: } \frac{\Omega_s}{\Omega_p} = \frac{\tan(0.25\pi)}{\tan(0.2\pi)} = \frac{1.0}{0.7265} = 1.3764. \quad N_{\text{IIR}} = \left\lceil \frac{\log_{10}[(10^5 - 1)/(10^{0.05} - 1)]}{2 \log_{10}(1.3764)} \right\rceil = \left\lceil \frac{\log_{10}(100000/0.122)}{2 \times 0.1387} \right\rceil = \left\lceil \frac{5.9136}{0.2774} \right\rceil = \mathbf{22} \rightarrow \mathbf{12} \text{ (BLT).}$$

- **Takeaway:** The IIR filter requires ≈ 6 biquad sections (12 poles), needing significantly fewer MAC operations per sample than the 60-tap FIR filter.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

- (a) Provide a detailed comparative analysis between FIR and IIR digital filters covering Phase Linearity, Stability, Computational Complexity, and Hardware Implementation. *(10 Marks)*
- (b) Under what practical engineering scenarios is an FIR filter mandatory over an IIR filter? *(5 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):** Comprehensive comparison matrix covering all 7 parameters listed in Section 2.2 *(10 Marks)*

- **Part (b):** 1. Digital Communications (preventing Intersymbol Interference), 2. High-Fidelity Audio processing (preserving transient phase coherence), 3. Biomedical ECG/EEG diagnostic monitoring (preventing distortion of QRS peak morphology) *(5 Marks)*
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5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

\title{\textbf{EE3621: Digital Signal Processing}} \\\Large Order comparison \\\large (III
N_fir, beta = signal.kaiserord(50, 0.1)
N_iir, Wn = signal.buttord(0.4, 0.5, 0.5, 50)
print(f"Required FIR Order: {N_fir}")
print(f"Required IIR Order: {N_iir}")
```